Kaustubh Hemant Sonawane

Bristol, UK | ksstubh.sonawane@gmail.com | +44 7388229743 | +91 8591441019 | www.github.com/Arishtanemi4 | www.linkedin.com/in/kaustubhson4 | https://kaustubh-sonawane-portfolio.vercel.app

UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Data Engineer with 3 years of industry experience designing and running data platforms in regulated financial services, plus an MSc in Data Science (University of Bristol). Designed a **Normalized SQL Schema** on **PostgreSQL** fed by scheduled **API** ingestion that merged in legacy data, built an **AWS** pipeline (**S3**, **Lambda**, **EventBridge**, **SageMaker**) that parses **300+** daily emails into structured records saving **60** analyst hours a month, and built the data and **RAG** layer for **Azure OpenAI** applications used by Compliance, Client Service and Investment. Delivered **Power BI** dashboards, implemented **RBAC** and network restrictions with the security team through to passing **VAPT**, and acted as de facto team lead for up to 4 interns. 3rd place at a Lloyds Bank-sponsored **fraud detection** datathon.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Data Architecture, Modelling & Pipelines

- Designed the **data model** and **Normalized SQL Schema** (**PostgreSQL**) for the internal analytics platform to the analysts' requirements, with **Python** ingestion scripts for the licensed **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly (fixed income) and monthly (equity); integrated the legacy analyst data previously held in Excel into the same database, with **Optimized SQL** queries, **Functions** and **Procedures** for efficient data insertion.
- Built a **Python ETL** pipeline that parses over **300** daily emails from credit rating agencies into structured data, **fuzzy-matches** company names against the ones analysts track and sends a daily consolidated email with an Excel copy stored in **S3**, **saving 60 analyst hours a month**; first built on **AWS Textract**, then moved to **Azure OpenAI** when an agency began sending emails as images; scheduled at 9 AM through **EventBridge**, **Lambda** and **AWS SageMaker**.
- Analyzed customer data with **Python Pandas** and **Market Basket Analysis** (**Apriori**) for the marketing team; migrated financial analysts from Excel to **Power BI** dashboards, improving efficiency by **10 hours a week**, and recreated them in the **ReactJS** frontend so the wider organization could use the data without Power BI licences.

Security, Governance & Cloud Infrastructure

- Designed the **AWS** infrastructure for the analytics platform and applications (**EC2**, **S3**, **ECR**, **Lambda**), provisioned **EC2** instances using **CloudFormation** templates prebuilt by the security team, and packaged the **FastAPI** microservices as **Docker** containers behind an **Nginx** reverse proxy.
- Automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins** and **GitLab CI/CD** pipeline.
- Built access control for sensitive financial data under security and compliance review: 3-tier **RBAC** with a **Python** authentication function against the company **LDAP** server, department-level data isolation, **AWS Security Groups** limiting access to company private IPs, and coordination with the security team on port whitelisting; the **Nginx** proxy was one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**.

AI-Enabling Data Platforms & Stakeholders

- Designed and built the data and retrieval layer for **LLM applications** end to end (**RAG** with **LlamaIndex** and **Azure OpenAI** embeddings, a separate **pgvector** database per department to keep each department's documents isolated, **Python FastAPI** microservices); **tool-calling** agents run **SQL** queries over the fund data to compare funds and generate PDF reports on request.
- Worked directly with Client Service, Compliance and Investment from requirements to release, with Compliance reviewing every application; built the first systems as the only engineer alongside my manager, later acted as de facto team lead for up to 4 interns; recognised with Best Team Project twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built; personal focus on the data preprocessing, scoring algorithm and RAG assistant. Preprocessing pipeline that joins 17 raw and augmented inputs (DepMap, GEO, HPA) into 14 clean tables (CSV and Parquet), with identity resolution across sources, duplicate and missing-value handling, integrity checks and a build manifest audit file; scoring ranks cancer cell lines by target inclusion and exclusion genes across six evidence layers; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Swindon Borough Council Fuel-Poverty Hack:** A household and neighborhood-based fuel poverty data analysis tool. | GitHub: https://github.com/Arishtanemi4/fuel-poverty.git

ACHIEVEMENTS

- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle **credit card fraud detection** problem.)
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git

SKILLS

- **Tools:** AWS (EC2, S3, ECR, Lambda, EventBridge, SageMaker, CloudFormation templates), Azure OpenAI, Power BI, Docker, Jenkins, GitLab (CI/CD), LlamaIndex, Nginx, Linux, LDAP, Git, VS Code, Claude Code.
- **Languages:** Python (pandas, FastAPI, Pydantic, scikit-learn, fuzzywuzzy), SQL, R (dplyr, tidyverse, ggplot2), JavaScript/TypeScript (NodeJS express, ReactJS), Shell
- **Databases:** PostgreSQL (schema design, SQL functions and procedures, pgvector), SQLite